

SALTED: A symmetry-adapted machine-learning program for predicting electron-densities in molecules and materials

Zekun Lou¹, Alan M. Lewis², Théophane Bernhard⁵, Lukas Seifert³, Agustin Salcedo⁶, Florian Kleemiss³, Mariana Rossi^{1,4}, and Andrea Grisafi^{5¶}

¹ MPI for the Structure and Dynamics of Matter, Hamburg, Germany ² Department of Chemistry, University of York, York, UK ³ Institute of Inorganic Chemistry, RWTH Aachen University, Landoltweg 1a, 52074 Aachen, Germany ⁴ Yusuf Hamied Department of Chemistry, Cambridge University, Cambridge, UK ⁵ Physicochimie des Électrolytes et Nanosystèmes Interfaciaux, Sorbonne Université, CNRS, F-75005 Paris, France ⁶ Laboratoire de physique de L'École normale supérieure de Paris, CNRS, ENS & Université PSL, Sorbonne Université, Université de Paris, F-75005 Paris, France ¶ Corresponding author

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#)
- [Repository](#)
- [Archive](#)

Editor: [↗](#)

Submitted: 02 September 2026

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](https://creativecommons.org/licenses/by/4.0/)).

Summary

SALTED provides an open-source Python package for machine learning the quantum-mechanical electron density, $n(\mathbf{r})$, in molecular and condensed-phase systems based on input atomic coordinates and species (Grisafi et al., 2019; Grisafi, Lewis, et al., 2023; Lewis et al., 2021). The program adopts a linear atom-centered decomposition of the electron density, which makes it highly transferable across diverse atomistic configurations sharing similar chemical environments. Because of this representation choice, SALTED is naturally interfaced with state-of-the-art electronic-structure programs based on atomic orbitals, namely CP2K (Iannuzzi et al., 2026), FHI-aims (Abbott et al., 2026), and PySCF (Sun et al., 2020), from which reference electron-density data can be generated and used to train a model. The learning algorithm is based on a symmetry-adapted extension of Gaussian process regression (Grisafi et al., 2018), making SALTED especially efficient in small-data regimes. Thanks to the implementation of vector-field kernel functions (Rossi et al., 2025), SALTED can also learn the first-order response of the electron density to applied electric fields, $\partial n(\mathbf{r})/\partial \mathbf{E}$. The application of SALTED within computational workflows has already shown its utility in a wide variety of contexts, including the calculation of polarization vectors (Grisafi, Bussy, et al., 2023) and polarizability tensors (Rossi et al., 2025), the accurate evaluation of Coulomb forces in QM/MM molecular-dynamics simulations (Grisafi & Salanne, 2024), and electronic-structure studies of large-scale 2D materials (Lou et al., 2026).

Statement of need

Electronic structure methods can be predictive for the simulation of the properties of a wide variety of materials. Among these methods, density-functional theory (DFT) has, perhaps, become the most successful and popular, having solved outstanding problems in areas as diverse as biochemistry, catalysis, nanotechnology and quantum materials (Burke, 2012; Jones, 2015). The popularity of DFT is based on its ability to provide sufficient accuracy for many practical applications at a modest cost. However, the cost of these calculations can only be considered modest in comparison to other, more advanced, electronic structure methods. For the current landscape of data-driven material science and molecular discovery, allied to the

quest of achieving first-principles accuracy for larger systems and longer time scales, even the cost of DFT calculations becomes prohibitively large.

A machine-learning method that can predict the most fundamental quantity of DFT, namely the real-space electronic density, therefore holds immense potential to deliver a single model that can be used to calculate a multitude of downstream material properties at a small fraction of the cost of DFT, while maintaining very similar accuracy. In this paper, we describe the SALTED software package, which allows electronic density and electronic-density response predictions over an atomic basis consistent with the underlying electronic-structure architectures.

The availability of this open-source software package provides the atomistic-simulation community with a practical and extensible framework for constructing machine-learning models of the electron density and its linear response from first-principles reference data. SALTED is intended for researchers in computational chemistry, condensed-matter physics, materials science, and molecular simulation who wish to accelerate electronic-structure calculations while retaining direct access to physically meaningful electronic observables. By interfacing with widely used electronic-structure packages, SALTED can be embedded into established computational workflows. The package is designed both for users seeking efficient predictions of electron densities and derived properties, and for method developers interested in extending the methodology or in interfacing density predictions with downstream electronic-structure and multiscale simulation tools. SALTED facilitates reproducible research and supports the development of machine-learning models for electronic structure by providing an open, documented, and reusable implementation of electron-density prediction.

State of the field

Two main directions have been independently followed to construct ML models of the electron density that either directly sample $n(\mathbf{r})$ on a real-space 3D grid, or expand $n(\mathbf{r})$ over a linear atomic basis. Examples of grid-based ML models with published open-source packages are: DeepDFT, based on equivariant graph neural networks (Jørgensen & Bhowmik, 2022); a linear-regression framework based on Jacobi-Legendre many-body descriptors (Focassio et al., 2023); Charge3net, based on a high-order equivariant neural network (Koker et al., 2024); and an extension of FIREANN, where an efficient sampling strategy is devised to drastically reduce the number of grid-point evaluations and to further treat the response to applied fields (Feng et al., 2025). Albeit not targeting directly the electronic density, the Materials Learning Algorithm (MALA) package learns the local density of states on a grid, and reconstructs $n(\mathbf{r})$ from this quantity (Cangi et al., 2025).

When compared with grid-based approaches, ML models that represent the density on an atomic basis carry the advantage of greatly reducing the amount of data at the price of an acceptably small error due to the finite expressiveness of the basis functions. The SALTED package is the result of one of the earliest ML approaches developed to learn the electron density via a set of atomic coefficients (Grisafi et al., 2019). Specifically, $n(\mathbf{r})$ is expanded over a linear basis made of radial functions R_n^λ and spherical harmonics Y_μ^λ centered around each atom of the system:

$$n(\mathbf{r}) = \sum_{in\lambda\mu} c_{in\lambda\mu} \sum_{\mathbf{u}} R_n^\lambda(|\mathbf{r} - \mathbf{r}_i - \mathbf{u}|) Y_\mu^\lambda(\widehat{\mathbf{r} - \mathbf{r}_i - \mathbf{u}})$$

where i are the atomic indexes, $\mathbf{u}$ are the cell translation vector when considering periodic systems (Lewis et al., 2021), and $c_{in\lambda\mu}$ are the density expansion coefficients. The reference data for $c_{in\lambda\mu}$ can directly be obtained from resolution of the identity (RI) methods implemented in electronic-structure codes that are based on atomic orbitals, making SALTED easy to interface with codes using such basis sets and methods.

87 As a key difference from the models previously mentioned, SALTED is based on a symmetry-
 88 adapted extension of Gaussian process regression (GPR). Spherical-tensor kernels (Grisafi
 89 et al., 2018), $\mathbf{K}^\lambda$, rather than equivariant neural-networks, are computed to satisfy exact
 90 $\mathcal{O}(3)$ symmetries (Grisafi et al., 2019), starting from atom-density features obtained using the
 91 featomic library (Bigi et al., 2026). While the adoption of a GPR method limits the scalability
 92 of SALTED over strongly heterogeneous datasets (Grisafi, Lewis, et al., 2023), the constrained
 93 function space defined by the kernels, together with the physically interpretable nature of
 94 the underlying atomistic features, helps avoid overfitting—thereby facilitating transferable
 95 extrapolations of the electron density across different system sizes. Moreover, when compared
 96 with data-hungry neural-network architectures, the data-efficiency of GPR-based approaches
 97 becomes a great advantage when training the model on high-level electronic-structure methods,
 98 for which only a small number of calculations ($\sim 10^{2-3}$) can be performed.

99 In addition, SALTED presents a couple of distinctive features. First, a suitable linear combination
 100 of the kernels $\mathbf{K}^\lambda$ allows SALTED to learn the first-order response of the density to uniform
 101 applied electric fields, $\partial n(\mathbf{r})/\partial E_k$, thereby ensuring the correct transformations of a 3D vector
 102 field (Rossi et al., 2025). Second, the use of the featomic library enables the inclusion of
 103 long-distance equivariants (LODE) features (Grisafi & Ceriotti, 2019), which are essential
 104 for learning the nonlocal redistribution of $n(\mathbf{r})$ in highly-polarizable systems such as metallic
 105 frameworks (Grisafi, Bussy, et al., 2023; Rossi et al., 2025).

106 We note that there are other ML packages that share the same philosophy as SALTED in
 107 representing the electron density similarly to the equation above, albeit with different features
 108 and capabilities: equivariant neural-network architectures based on the E3NN model (Rackers
 109 et al., 2023); scdp, which augments the atom-centered basis with additional “virtual” orbitals
 110 placed at non-atomic sites (e.g., bond midpoints) to improve expressivity, and regresses the
 111 expansion coefficients with a high-capacity equivariant neural network (Fu et al., 2024);
 112 and lastly, an equivariant neural-network architecture that predicts atom-centered density
 113 coefficients as an intermediate representation for learning energies and forces (Bogojeski et al.,
 114 2026).

115 Software design

116 SALTED is organized as a workflow in three stages: dataset preparation, model training, and
 117 prediction (Figure 1). Each stage is configured through a dedicated section of a single `inp.yaml`
 118 file, and each step within a stage is exposed as an independently invocable command, `python`
 119 `-m salted.<step>`. GitHub Continuous Integration (CI) is enabled for testing the package:
 120 unit tests cover the utility modules, and pipeline tests run the full workflow through every
 121 electronic-structure interface.

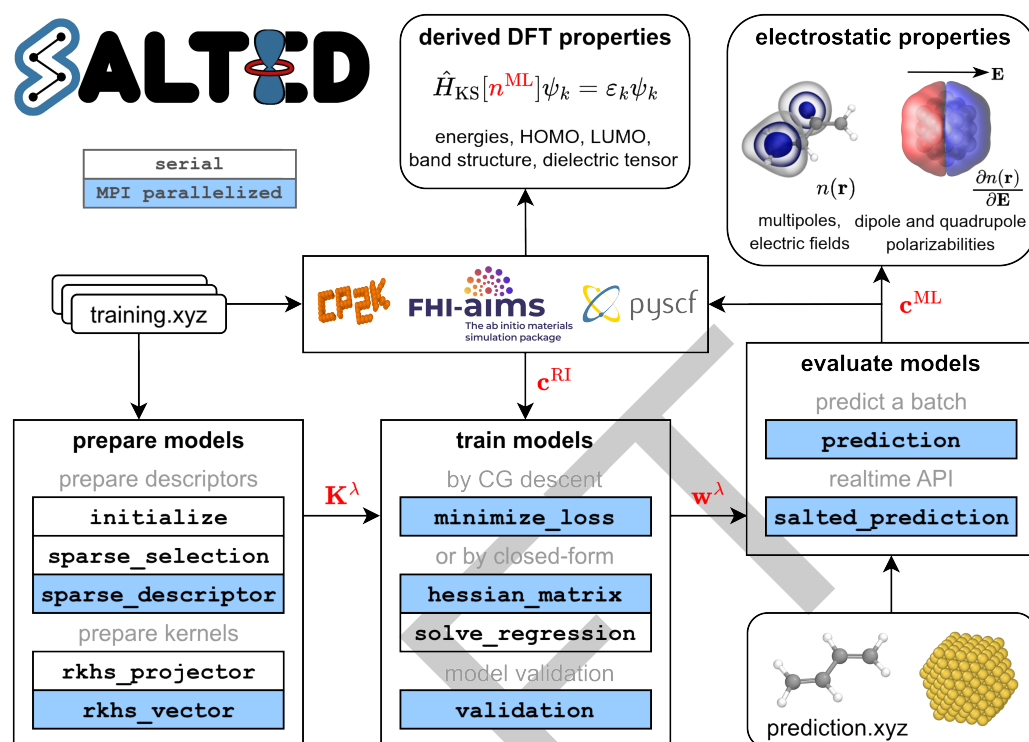


Figure 1: SALTED workflow. Reference density or density-response RI coefficients c^{RI} of the training dataset are generated by CP2K, FHI-aims, or PySCF. Model preparation builds λ -descriptors and assembles kernels K^λ of the training structures, with optional sparsification. Model training obtains regression weights w^λ using c^{RI} and K^λ . The trained models can be quickly validated to determine their accuracy, before being used to evaluate unseen structures. Predicted RI coefficients c^{ML} are used either directly, to evaluate electrostatic properties such as multipoles, electric fields, and polarizabilities, or as an input density from which the Kohn–Sham Hamiltonian is built and further obtain DFT properties after one diagonalization step. Shaded boxes mark stages that are parallelized by MPI.

This step-wise organization allows a run to be check-pointed, restarted, and distributed across separate HPC allocations, with the intermediate quantities of each step available on disk. Such control is necessary in practice: training a capable model over a large chemical space or on condensed-phase systems usually requires many HPC nodes, and the most efficient parallelization strategy differs from step to step. Users can resume an interrupted run, reuse descriptors across hyperparameter choices, and inspect intermediate quantities to improve models. We accept a less immediate interactive experience that a single-call library API would provide in exchange for workflows that remain manageable by the user.

SALTED integrates with a variety of electronic-structure codes based on atomic orbitals, CP2K (Iannuzzi et al., 2026), FHI-aims (Abbott et al., 2026), and PySCF (Sun et al., 2020), thus supporting a large fraction of community-developed electronic-structure software. Density data from CP2K are provided on uniform grids as *.cube files following pseudo-valence representations of $n(\mathbf{r})$; SALTED can then compute the required RI coefficients via a dedicated density_fitting module. FHI-aims is an all-electron electronic-structure program; in this case, density data for isolated and periodic systems are directly obtained as RI coefficients on the same footing. Finally, pseudo-valence PySCF density data for finite molecular systems can be obtained from SALTED by calling relevant PySCF functions.

SALTED unifies the training data obtained from these different codes within a single shared format. This requires extra engineering in the interfaces, but it lets research groups keep the electronic-structure code they already use in production, together with its established settings, and integrate SALTED into an existing workflow rather than switching electronic-

143 structure codes. CI tests cover all the interfaces. We note that both CP2K and PySCF work
144 with Gaussian-type orbitals, enabling straightforward analytical calculations of density and
145 density-response derived properties from the predicted SALTED coefficients, such as Hartree
146 energies, electric-fields, dipoles and polarizabilities. While some of these properties are already
147 computed and printed by SALTED upon density prediction, others could readily be implemented
148 as dedicated post-processing modules.

149 The performance-critical steps, including equivariant descriptor contraction and the construction
150 of Hessian matrix, are implemented as numba-compiled Python kernels (Lam et al., 2015). The
151 numba kernels are carefully composed and benchmarked, and their portability does not come at
152 the cost of performance. MPI parallelization is optional: parallelizable steps run serially by
153 default and use mpi4py (Dalcin & Fang, 2021) automatically when launched under mpirun.
154 OpenMP and Python thread pools would parallelize these steps within a single shared-memory
155 node, whereas MPI covers the full range of hardware of the users, from personal computers and
156 small workstations to multi-node HPC clusters. To improve portability, the supplied Dockerfile
157 can be used to build a self-contained image with an MPI-enabled version of SALTED.

158 Research impact statement

159 The integration of SALTED predictions within computational workflows has already been shown
160 useful in real-case scientific applications. For example, SALTED predictions of the charge-density
161 response in metallic electrodes could be used to achieve a $\sim 10^3$ speedup in the calculation of
162 DFT-level electrostatic forces entering the quantum-mechanics/molecular-mechanics simulation
163 of model ionic capacitors (Grisafi, Lewis, et al., 2023). More recently, the use of the predicted
164 density as a DFT input to perform a single Kohn-Sham Hamiltonian diagonalization has enabled
165 the study of the electronic properties of twisted bilayer Moiré materials over extremely large
166 supercells (Lou et al., 2026). Beyond electrostatic and DFT properties, SALTED enables the
167 calculation of electron-density-based features at scale. A relevant example is the application
168 to quantum-chemistry molecular interaction analysis (Fabrizio et al., 2020). Additionally, the
169 RI density coefficients are directly applicable to crystal structure refinement from X-ray and
170 electron diffraction data (Seifert et al., 2026), making the integration of SALTED predictions
171 within these algorithms highly promising.

172 AI usage disclosure

173 We declare that no AI tool was used to make core code design decisions, nor were they used
174 to make scientific decisions. Generative AI tools were employed to help with code refactoring
175 and cleaning and with text polishing.

176 Acknowledgements

177 We acknowledge funding from the French National Research Agency under the France 2030
178 program (PEPR BATMAN, Grant No. ANR-22-PEBA-0002 and PEPR DIADEM, Grant
179 No. ANR-22-PEXD-0001) and the German Research Foundation (DFG) Project-ID 555467911
180 - CRC 1772 / TP A06.

181 References

182 Abbott, J. W., Mera Acosta, C., Akkoush, A., Ambrosetti, A., Atalla, V., Bagrets, A., Behler,
183 J., Berger, D., Bertschi, H., Bieniek, B., Björk, J., Blum, V., Bohloul, S., Box, C. L., Boyer,
184 N. J., Brambila, D. S., Bramley, G. A., Bryenton, K. R., Camarasa-Gómez, M., ... Zhu, T.
185 (2026). Roadmap on advancements of the FHI-aims software package. *Electron. Struct.*
186 <https://doi.org/10.1088/2516-1075/ae8067>

- Bigi, F., Abbott, J. W., Loche, P., Mazitov, A., Tisi, D., Langer, M. F., Goscinski, A., Pegolo, P., Chong, S., Goswami, R., Febrer, P., Chorna, S., Kellner, M., Ceriotti, M., & Fraux, G. (2026). Metatensor and metatomic: Foundational libraries for interoperable atomistic machine learning. *J. Chem. Phys.*, 164(6), 064113. <https://doi.org/10.1063/5.0304911>
- Bogojeski, M., Hasyim, M. R., Vogt-Maranto, L., Müller, K.-R., Burke, K., & Tuckerman, M. E. (2026). *Enhancing molecular dynamics with equivariant machine-learned densities*. <https://doi.org/10.48550/arXiv.2604.24563>
- Burke, K. (2012). Perspective on density functional theory. *J. Chem. Phys.*, 136(15), 150901. <https://doi.org/10.1063/1.4704546>
- Cangi, A., Fiedler, L., Brzozza, B., Shah, K., Callow, T. J., Kotik, D., Schmerler, S., Barry, M. C., Goff, J. M., Rohskopf, A., Vogel, D. J., Modine, N., Thompson, A. P., & Rajamanickam, S. (2025). Materials learning algorithms (MALA): Scalable machine learning for electronic structure calculations in large-scale atomistic simulations. *Comput. Phys. Commun.*, 314, 109654. <https://doi.org/10.1016/j.cpc.2025.109654>
- Dalcin, L., & Fang, Y.-L. L. (2021). mpi4py: Status update after 12 years of development. *Comput. Sci. Eng.*, 23(4), 47–54. <https://doi.org/10.1109/MCSE.2021.3083216>
- Fabrizio, A., Briling, K., Grisafi, A., & Corminboeuf, C. (2020). Learning (from) the electron density: Transferability, conformational and chemical diversity. *Chimia*, 74(4), 232–236. <https://doi.org/10.2533/chimia.2020.232>
- Feng, C., Zhang, Y., & Jiang, B. (2025). Efficient sampling for machine learning electron density and its response in real space. *J. Chem. Theory Comput.*, 21(2), 691–702. <https://doi.org/10.1021/acs.jctc.4c01355>
- Focassio, B., Domina, M., Patil, U., Fazzio, A., & Sanvito, S. (2023). Linear jacobi-legendre expansion of the charge density for machine learning-accelerated electronic structure calculations. *Npj Comput. Mater.*, 9(1), 87. <https://doi.org/10.1038/s41524-023-01053-0>
- Fu, X., Rosen, A., Bystrom, K., Wang, R., Musaelian, A., Kozinsky, B., Smidt, T., & Jaakkola, T. (2024). A recipe for charge density prediction. *Advances in Neural Information Processing Systems (NeurIPS)*. <https://doi.org/10.52202/079017-0310>
- Grisafi, A., Bussy, A., Salanne, M., & Vuilleumier, R. (2023). Predicting the charge density response in metal electrodes. *Phys. Rev. Mater.*, 7(12), 125403. <https://doi.org/10.1103/PhysRevMaterials.7.125403>
- Grisafi, A., & Ceriotti, M. (2019). Incorporating long-range physics in atomic-scale machine learning. *J. Chem. Phys.*, 151(20), 204105. <https://doi.org/10.1063/1.5128375>
- Grisafi, A., Fabrizio, A., Meyer, B., Wilkins, D. M., Corminboeuf, C., & Ceriotti, M. (2019). Transferable machine-learning model of the electron density. *ACS Cent. Sci.*, 5(1), 57–64. <https://doi.org/10.1021/acscentsci.8b00551>
- Grisafi, A., Lewis, A. M., Rossi, M., & Ceriotti, M. (2023). Electronic-structure properties from atom-centered predictions of the electron density. *J. Chem. Theory Comput.*, 19(14), 4451–4460. <https://doi.org/10.1021/acs.jctc.2c00850>
- Grisafi, A., & Salanne, M. (2024). Accelerating QM/MM simulations of electrochemical interfaces through machine learning of electronic charge densities. *J. Chem. Phys.*, 161(2), 024109. <https://doi.org/10.1063/5.0218379>
- Grisafi, A., Wilkins, D. M., Csányi, G., & Ceriotti, M. (2018). Symmetry-adapted machine learning for tensorial properties of atomistic systems. *Phys. Rev. Lett.*, 120(3), 036002. <https://doi.org/10.1103/PhysRevLett.120.036002>
- Iannuzzi, M., Wilhelm, J., Stein, F., Bussy, A., Elgabarty, H., Golze, D., Hehn, A.-S., Graml, M., Marek, S., Gökmen, B. S., Schran, C., Forbert, H., Khaliullin, R. Z., Kozhevnikov,

- 234 A., Taillefumier, M., Meli, R., Rybkin, V. V., Brehm, M., Schade, R., ... Kühne, T. D.
235 (2026). The CP2K program package made simple. *J. Phys. Chem. B*, 130(4), 1237–1310.
236 <https://doi.org/10.1021/acs.jpcb.5c05851>
- 237 Jones, R. O. (2015). Density functional theory: Its origins, rise to prominence, and future.
238 *Rev. Mod. Phys.*, 87(3), 897–923. <https://doi.org/10.1103/RevModPhys.87.897>
- 239 Jørgensen, P. B., & Bhowmik, A. (2022). Equivariant graph neural networks for fast electron
240 density estimation of molecules, liquids, and solids. *Npj Comput. Mater.*, 8(1), 183.
241 <https://doi.org/10.1038/s41524-022-00863-y>
- 242 Koker, T., Quigley, K., Taw, E., Tibbetts, K., & Li, L. (2024). Higher-order equivariant neural
243 networks for charge density prediction in materials. *Npj Comput. Mater.*, 10(1), 161.
244 <https://doi.org/10.1038/s41524-024-01343-1>
- 245 Lam, S. K., Pitrou, A., & Seibert, S. (2015). Numba: A LLVM-based python JIT compiler.
246 *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC*, 7.
247 <https://doi.org/10.1145/2833157.2833162>
- 248 Lewis, A. M., Grisafi, A., Ceriotti, M., & Rossi, M. (2021). Learning electron densities in the
249 condensed phase. *J. Chem. Theory Comput.*, 17(11), 7203–7214. <https://doi.org/10.1021/acs.jctc.1c00576>
- 250
- 251 Lou, Z., Lewis, A. M., & Rossi, M. (2026). Long-range machine learning of electron density
252 for twisted bilayer moiré materials. *PRX Intelligence*, 1(1), 013015. <https://doi.org/10.1103/4575-9cmx>
- 253
- 254 Rackers, J. A., Tecot, L., Geiger, M., & Smidt, T. E. (2023). A recipe for cracking the
255 quantum scaling limit with machine learned electron densities. *Machine Learning: Sci.*
256 *Technol.*, 4(1), 015027. <https://doi.org/10.1088/2632-2153/acb314>
- 257 Rossi, M., Rossi, K., Lewis, A. M., Salanne, M., & Grisafi, A. (2025). Learning the electrostatic
258 response of the electron density through a symmetry-adapted vector field model. *J. Phys.*
259 *Chem. Lett.*, 16(9), 2326–2332. <https://doi.org/10.1021/acs.jpcllett.5c00165>
- 260 Seifert, L. M., Brück, D., Piel, T. M., & Kleemiss, F. (2026). Grid-free quantum crystallographic
261 refinement using unbiased analytic atomic form factors from density fitting. *Z. Kristallogr.*
262 *- Cryst. Mater.* <https://doi.org/10.1515/zkri-2026-0013>
- 263 Sun, Q., Zhang, X., Banerjee, S., Bao, P., Barbry, M., Blunt, N. S., Bogdanov, N. A.,
264 Booth, G. H., Chen, J., Cui, Z.-H., Eriksen, J. J., Gao, Y., Guo, S., Hermann, J.,
265 Hermes, M. R., Koh, K., Koval, P., Lehtola, S., Li, Z., ... Chan, G. K.-L. (2020). Recent
266 developments in the PySCF program package. *J. Chem. Phys.*, 153(2), 024109. <https://doi.org/10.1063/5.0006074>
- 267